Project: Worked examples
Title: Pipe expansion, loops and anchor forces (screening)
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1. Scope and method

Calculates the thermal growth of a pipe run between anchors, the leg or loop that absorbs it, and the force and moment on the anchors. It is a screening calculation, not a piping stress analysis.

WHAT IT CALCULATES	RELATION
Pipe section	$d_i = OD - 2 \cdot t$; $A_m = \pi/4 \cdot (OD^2 - d_i^2)$; $I = \pi/64 \cdot (OD^4 - d_i^4)$; $Z = I/(OD/2)$
Temperature change and growth	$\Delta T = T_{operate} - T_{install}$, or as entered; L_{tot} = L, or L + L2 for a Z-bend; $\Delta L = \alpha \cdot L_{tot} \cdot \lvert \Delta T \rvert$, its sign giving the direction
Fully restrained pipe	$\sigma_r = E \cdot \alpha \cdot \lvert \Delta T \rvert$ (E at the installation temperature); $F_r = E_r \cdot \alpha \cdot \lvert \Delta T \rvert \cdot A_m$ ($E_r = E_{hot}$, else E)
Guided cantilever, required leg	Δ = ΔL for an offset leg or Z-bend and $\Delta L/2$ for a U-loop; $L_{req} = \sqrt{(3 \cdot E \cdot OD \cdot \Delta / Sa)}$; PASS when $L_{req} \le$ the available leg H
Force and moment of the available leg	$F_{leg} = 12 \cdot E_r \cdot I \cdot \Delta / H^3$ (along the run); $M_{leg} = 6 \cdot E_r \cdot I \cdot \Delta / H^2$ (= $F_{leg} \cdot H/2$); $\sigma_{leg} = 3 \cdot E \cdot OD \cdot \Delta / H^2 = Sa \cdot (L_{req}/H)^2$
Runs of an offset leg or Z-bend	the runs bend for the leg's growth $\Delta_{leg} = \alpha \cdot \lvert \Delta T \rvert \cdot H$ under one force $F_{run} = 12 \cdot E_r \cdot I \cdot \Delta_{leg} / (L^3 + L2^3)$ along the leg (L2 = 0 for an L-bend); run i takes $L_i^3/(L^3 + L2^3)$ of Δ_{leg}, with $\sigma_i = 3 \cdot E \cdot OD \cdot \Delta_i / L_i^2$ and $M_i = F_{run} \cdot L_i/2$, and passes when $\sqrt{(\sigma_i/Sa)} \le 1$
Force on the anchors	$\sqrt{(F_{leg}^2 + F_{run}^2)}$ from the two perpendicular components; the moment reported is the largest root moment of the legs and runs
Screening and bellows	$OD \cdot Y/(L_{dev} - U)^2 \le K1$ with $Y = \alpha \cdot \lvert \Delta T \rvert \cdot U$ (offset leg or Z-bend $L_{dev} = L_{tot} + H$, $U = \sqrt{(L_{tot}^2 + H^2)}$; U-loop $L_{dev} = L + 2 \cdot H$, $U = L$); axial bellows anchor force = $P \cdot A_{eff} + k \cdot \Delta L$, with no leg check and no screening

Basis: Process piping — flexibility analysis, exemption from formal analysis[1](cl. 319.4.1 (screening relation; number to be checked against the owner's copy)); Process piping — displacement stress range and reactions[1](cl. 319.3.2 and 319.5 (reference and hot moduli; numbers to be checked against the owner's copy)); Metallic industrial piping — flexibility analysis[2](cl. 12 (number to be checked against the owner's copy)).

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
layout	What absorbs the movement	U-loop (two legs, each takes half the movement)		drawing: DRG-M-210 rev B	any

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
$temp_{mode}$	Temperature change from	install and operating temperatures		assumption: Install and operating temperatures are both known.	any
OD	Pipe outside diameter	88.9	mm	drawing: DRG-M-210 rev B	10 … 1500 mm
t_{wall}	Pipe wall thickness	3.2	mm	drawing: DRG-M-210 rev B	0.5 … 60 mm
L	Anchor spacing: the straight length that grows (an offset leg or Z-bend: the first run, which also bends for the leg's growth)	50	m	drawing: DRG-M-210 rev B	1 … 500 m
H	Available leg length (each leg of a U-loop; an offset leg: the leg off the run, which may be the longer of the two); optional for an axial bellows, which has no leg; no default	4	m	drawing: DRG-M-210 rev B	0.5 … 50 m
E	Young's modulus of the pipe material at the installation (reference) temperature, for the stress check, in GPa (steel 210, copper 120, PE and PP about 1; not MPa)	210	GPa	assumption: W3 seed 'Carbon steel (generic)', indicative; replace with the project's material record.	0.5 … 500 GPa
α	Mean coefficient of thermal expansion between the install and operating temperatures	12	um/(m·K)	assumption: W3 seed 'Carbon steel (generic)' (12e-6 1/K), indicative; replace with the project's material record.	1 … 250 um/(m·K)
$T_{install}$	Temperature at installation (when the run is anchored)	10	°C	assumption: Mean site temperature when the main is anchored.	-50 … 50 °C
$T_{operate}$	Operating temperature (hot or cold)	80	°C	spec: SPEC-M-04 §3: LTHW flow temperature	-50 … 500 °C
Sa	Allowable displacement-stress range for the legs and runs (your piping code and its formula; optional for an axial bellows; no default)	100	MPa	assumption: Invented illustrative allowable range; take Sa from the project's piping code and its formula.	5 … 500 MPa

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
K_{scr}	Screening limit K1 for OD·Y/(L − U)², in mm²/m² (or in²/ft², ...) from your piping code (optional; blank = screening not evaluated, so the sheet does not pass; no default)	120	mm²/m²	assumption: Invented illustrative screening limit; take K1 from the project's piping code.	1 ... 1000 mm²/m²

3. Assumptions

A1. The result is a screen, not a piping stress analysis. (Stated scope.)
A2. The result tells whether a simple leg or loop is plausible and what it puts on its anchors. (Stated scope.)
A3. It does not replace a flexibility analysis of the real routing, including elbow flexibility, intensification, torsion, branches, supports and loads. (Stated scope.)
A4. A guided cantilever is straight, uniform, fixed against rotation at both ends and loaded only by the movement it absorbs. (Beam theory.)
A5. Its stiffness is $12 \cdot E \cdot I / H^3$ and its end moment is $F \cdot H / 2$. (Beam theory.)
A6. Elbows are taken as rigid and the leg as slender. (Beam theory.)
A7. The beam model is not valid for legs shorter than about ten diameters. (Beam theory.)
A8. No stress-intensification factor and no elbow flexibility are applied. (Stated limit.)
A9. The stress at the elbow root of a real bend is higher than the beam stress M/Z. (Stated limit.)
A10. A leg that passes here can still exceed Sa at the elbow, which is why the result is a screen. (Stated limit.)
A11. A U-loop is two identical legs in series, each a guided cantilever taking half the growth. (Loop model.)
A12. The cross leg of a U-loop is a rigid guide. (Loop model.)
A13. The force along the line is the same in both legs of a U-loop. (Loop model.)
A14. The moment M is the moment at the leg roots, the corners of the loop. (Loop model.)
A15. It reaches an anchor only where the leg ends at the anchor, as in an offset leg. (Loop model.)
A16. The growth of the loop's own legs, which the cross leg has to absorb, is not checked. (Loop model.)
A17. The run of an offset leg bends for the leg's growth as a second guided cantilever, checked against Sa. (Offset-leg and Z-bend model.)
A18. In a Z-bend the two runs are in series under one force. (Offset-leg and Z-bend model.)
A19. Each run takes its share of the leg's growth in proportion to L_i^3 and is checked as its own cantilever. (Offset-leg and Z-bend model.)
A20. The leg of a Z-bend takes the growth of both runs. (Offset-leg and Z-bend model.)
A21. The force on the anchors is the resultant of the two perpendicular forces along the run and along the leg. (Offset-leg and Z-bend model.)
A22. $anchor_{moment}$ is the largest root moment of the legs and runs, $M = F \cdot L / 2$ with each end fixed against rotation. (Stated limit.)
A23. It is not the moment on a frame anchor, which depends on the whole frame. (Stated limit.)
A24. A formal analysis is needed before an anchor is sized for moment. (Stated limit.)
A25. α is the mean coefficient of expansion between the install and operating temperatures. (Material library.)
A26. E is the modulus at the installation temperature, used for the stress check as the piping codes do. (The ASME process piping code, cl. 319.3.2 and 319.5 (numbers to be checked against the owner's copy).)
A27. E_{hot}, when given, is used only for the reactions on the anchors. (Code convention to be checked.)
A28. The material properties are taken from a library material applied at the right temperature. (Material library.)
A29. Sa and K1 are the owner's records from the owner's own copy of the piping code or the design basis. (No table from a standard is shipped.)
A30. No allowable stress, K1 or code constant ships. (No table from a standard is shipped.)
A31. The sheet applies the relation to the numbers entered, whatever code or load case they come from.

(No table from a standard is shipped.)
A32. No weight, spring-hanger or cold-spring effect is included in the leg force. (Stated limit.)
A33. The anchor force adds support friction μ·w·L only when μ and the weight per metre are entered. (Stated limit; a warning says so when they are not.)
A34. The bellows anchor force is the pressure thrust $P \cdot A_{eff}$ plus the spring force $k \cdot \Delta L$ at the governing pressure. (Stated limit.)
A35. The governing pressure is the design or test pressure, whichever governs, because compression and extension ratings can differ. (Stated limit.)
A36. Guide friction and any other bellows load come from the manufacturer's data. (Stated limit.)
A37. $temp_{mode}$ (Temperature change from) is taken as install and operating temperatures. (Install and operating temperatures are both known.)
A38. E (Young's modulus of the pipe material at the installation (reference) temperature, for the stress check, in GPa (steel 210, copper 120, PE and PP about 1; not MPa)) is taken as 210 GPa. (W3 seed 'Carbon steel (generic)', indicative; replace with the project's material record.)
A39. α (Mean coefficient of thermal expansion between the install and operating temperatures) is taken as 12 um/(m·K). (W3 seed 'Carbon steel (generic)' (12e-6 1/K), indicative; replace with the project's material record.)
A40. $T_{install}$ (Temperature at installation (when the run is anchored)) is taken as 10 °C. (Mean site temperature when the main is anchored.)
A41. Sa (Allowable displacement-stress range for the legs and runs (your piping code and its formula; optional for an axial bellows; no default)) is taken as 100 MPa. (Invented illustrative allowable range; take Sa from the project's piping code and its formula.)
A42. K_{scr} (Screening limit K1 for OD·Y/(L – U)², in mm²/m² (or in²/ft², ...) from your piping code (optional; blank = screening not evaluated, so the sheet does not pass; no default)) is taken as 120 mm²/m². (Invented illustrative screening limit; take K1 from the project's piping code.)
A43. Layout: the U-loop is two legs of 4 m in series, each a guided cantilever taking half the movement (21 mm); the cross leg is a rigid guide and its width does not enter. Screening only — not a piping stress analysis. (Guided-cantilever model.)

4. Working

[9] $dT = T_{operate} - T_{install}$
$= 80\ degC - 10\ degC$
$= 70\ K$
Temperature change, operating – install
[11] $E_r = E$
$= 210\ GPa$
$= 210\ GPa$
Modulus for the reactions (E, no E_{hot} given)
[12] $D_i = OD - 2 * t_{wall}$
$= 88.9\ mm - 2 \times 3.2\ mm$
$= 82.5\ mm$
Inside diameter
[13] $A_m = pi / 4 * (OD^2 - D_i^2)$
$= \pi / 4 \times ((88.9\ mm)^2 - (82.5\ mm)^2)$
$= 861.55\ mm^2$
Metal cross-section
[14] $I_x = pi / 64 * (OD^4 - D_i^4)$
$= \pi / 64 \times ((88.9\ mm)^4 - (82.5\ mm)^4)$
$= 79.206\ cm^4$
Second moment of area of the pipe section
[15] $Z_x = I_x / (OD / 2)$
$= 79.206\ cm^4 / (88.9\ mm / 2)$
$= 17.819\ cm^3$

Section modulus

[16] L_{tot} = L
= 50 m
= 50 m

Length that grows between the anchors

[17] dL = α * L_{tot} * abs(dT)
= 12 um/(m·K) × 50 m × abs(70 K)
= 42 mm

Thermal growth between the anchors, as a magnitude

[18] dL_{signed} = α * L_{tot} * dT
= 12 um/(m·K) × 50 m × 70 K
= 42 mm

Thermal movement with its sign: positive lengthens, negative shortens

[19] σ_{res} = E * α * abs(dT)
= 210 GPa × 12 um/(m·K) × abs(70 K)
= 176.4 MPa

Stress in the fully restrained pipe (E at the installation temperature)

[20] F_{res} = E_r * α * abs(dT) * A_m
= 210 GPa × 12 um/(m·K) × abs(70 K) × 861.55 mm²
= 151.98 kN

Force on the anchors of the fully restrained pipe

[21] $F_{friction}$ = support friction not entered
= μ and w_{pipe} blank
= 0 kN

Friction of the sliding supports (not entered: none in the anchor force)

[24] u_{res} = σ_{res} / Sa
= 176.4 MPa / 100 MPa
= 1.764

Fully restrained stress as a fraction of Sa (above 1: the movement has to be absorbed)

[25] Δ = dL / 2
= 42 mm / 2
= 21 mm

Movement each leg of the U-loop absorbs (half of it)

[26] L_{req} = sqrt(3 * E * OD * Δ / Sa)
= sqrt(3 × 210 GPa × 88.9 mm × 21 mm / 100 MPa)
= 3.4295 m

Leg length at which the guided-cantilever bending stress reaches Sa

[27] u_{leg} = L_{req} / H
= 3.4295 m / 4 m
= 0.85737

Required leg over available leg

[28] F_{leg} = 12 * E_r * I_x * Δ / H^3
= 12 × 210 GPa × 79.206 cm⁴ × 21 mm / (4 m)^3
= 0.65493 kN

Force of the available leg on its anchors or guides

[29] M_{leg} = 6 * E_r * I_x * Δ / H^2
= 6 × 210 GPa × 79.206 cm⁴ × 21 mm / (4 m)^2
= 1.3099 kN·m

Moment at the root of the available leg

[30] σ_{leg} = 3 * E * OD * Δ / H^2
= 3 × 210 GPa × 88.9 mm × 21 mm / (4 m)^2
= 73.509 MPa

Bending stress at the root of the available leg (E at the installation temperature)

[31] $u_{sigma} = \sigma_{leg} / Sa$
= 73.509 MPa / 100 MPa
= 0.73509
Bending stress over Sa (= u_{leg}^2)
[32] $F_{along,run} = F_{leg} + F_{friction}$
= 0.65493 kN + 0 kN
= 0.65493 kN
Component of the anchor force along the run: the loop's force plus the support friction
[33] $F_{along,leg}$ = no component along the leg
= a U-loop is in line
= 0 kN
Component of the anchor force along the leg: none for a U-loop
[34] $anchor_{force} = F_{along,run}$
= 0.65493 kN
= 0.65493 kN
Force on the anchors (the force along the loop; support friction only if μ and w_{pipe} are entered)
[35] $anchor_{moment} = M_{leg}$
= 1.3099 kN·m
= 1.3099 kN·m
Moment at the root of the loop's legs
[36] $L_{dev} = L + 2 * H$
= 50 m + 2 × 4 m
= 58 m
Developed length between the anchors, run plus two legs
[37] $U_a = L$
= 50 m
= 50 m
Straight distance between the anchors (the loop is in line)
[38] $Y_{scr} = \alpha * abs(dT) * U_a$
= 12 um/(m·K) × abs(70 K) × 50 m
= 42 mm
Resultant displacement Y: the free growth between the anchors
[39] $scr = OD * Y_{scr} / (L_{dev} - U_a)^2$
= 88.9 mm × 42 mm / (58 m − 50 m)^2
= 58.341 mm²/m²
Screening ratio OD·Y/(L − U)²
[41] $u_{scr} = scr / K_{scr}$
= 58.341 mm²/m² / 120 mm²/m²
= 0.48617
Screening ratio over the K1 record
[42] pass = K1 given and every check passes
= 2 check(s)
= 1
1 when the leg, the run (offset leg) and the screening checks all pass; 0 when one fails or when no K1 was entered

5. Results

SYMBOL	VALUE	DESCRIPTION
$anchor_{force}$	0.6549 kN	Force on the anchors (resultant)
$anchor_{moment}$	1.310 kN·m	Largest root moment (guided-cantilever model)
dL	42 mm	Thermal movement between the anchors (magnitude)
D_i	82.50 mm	Inside diameter
A_m	861.6 mm²	Metal cross-section

SYMBOL	VALUE	DESCRIPTION
I_x	79.21 cm⁴	Second moment of area
Z_x	17.82 cm³	Section modulus
dT	70 K	Temperature change
L_{tot}	50 m	Length that grows
dL_{signed}	42 mm	Thermal movement, signed
σ_{res}	176.4 MPa	Fully restrained stress
F_{res}	151.98 kN	Fully restrained force
direction	"growth: the pipe lengthens; fully restrained it presses on its anchors (compression)"	Direction of the movement
$F_{friction}$	0 kN	Support friction on the run
Δ	21 mm	Movement absorbed by each leg
L_{req}	3.429 m	Required leg length (governing)
F_{leg}	0.6549 kN	Force of the leg
M_{leg}	1.310 kN·m	Moment at the leg root
σ_{leg}	73.51 MPa	Bending stress at the leg root
$F_{along,run}$	0.6549 kN	Anchor force component along the run
$F_{along,leg}$	0 kN	Anchor force component along the leg
L_{dev}	58 m	Developed length
U_a	50 m	Anchor distance
Y_{scr}	42 mm	Resultant displacement Y
scr	58.341 mm²/m²	Screening ratio OD·Y/(L − U)²
screen	"PASS"	Screening result
pass	1	All checks pass
u_{res}	1.77	Fully restrained stress over Sa
u_{leg}	0.86	Required leg over available leg
u_{sigma}	0.74	Bending stress over Sa
u_{scr}	0.49	Screening ratio over K1

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Available leg against the required leg	3.429 m	4 m	0.86	

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Screening ratio against the limit K1 (not met: the line needs a formal stress analysis)	58.341 mm²/m²	120 mm²/m²	0.49	[1](cl. 319.4.1 (screening form; number to be checked against the owner's copy))

7. Applicability

WARNING records: tabulated values not taken from a reference record are E, α (Material properties); OD, t_{wall} (Pipe section (outside diameter and wall)); Sa (Allowable displacement-stress range); K_{scr} (Screening limit K1). Enter each as one so its source, edition and licence travel with it. [record]
WARNING μ: the anchor force holds no support friction. Enter μ and the weight per metre, with the length on slides, to include it. [friction]

8. Verification

Template bs.pipe-expansion@1.5.1 Content hash 29443308b79dc6fc Engine calc-engine/1.0.0
Known-answer cases passed 5/5 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

9. References

1. ASME B31.3-2022.
2. BS EN 13480-3:2017.